

Conditional Layer Invariance in Protein Language Models: Position Leakage and Readout Choice Determine the Apparent Benefit of Multi-Scale Embeddings

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Abstract

Motivation: Protein language models such as ESM-2 encode biochemical information across 33 transformer layers. Multi-layer embedding fusion is often used under the assumption that different layers provide complementary mutation-relevant information. However, the conditions under which this assumption holds for mutation effect prediction remain poorly characterized. **Results:** We benchmarked two 650M-parameter PLMs on the ProteinGym substitution benchmark using per-layer probing under three feature arms and four train–test splits, multi-scale pooling, logit lens scoring, layer-wise similarity analysis, and a position-lookup control. Representational geometry changed substantially across depth (adjacent-layer cosine ≈ 0.98), yet for SaProt the linear CKA between layers 0 and 32 remained 0.81 despite a raw cosine of -0.01 : the network rotates its representations while preserving linearly decodable information. Under the standard random split, all 33 layers again gave nearly indistinguishable predictions ($\rho \approx 0.60$), but a control regressor given only random position vectors reproduced this level ($\rho = 0.600$) and collapsed to 0.10–0.14 under leakage-free splits (paired drop -0.46 , Wilcoxon $p = 5.2 \times 10^{-12}$): random-split layer invariance is largely position memorization. Under three leakage-free splits (modulo, contiguous, position-grouped), mutation-specific embeddings reached ρ up to 0.53, the best layers were consistently late (layers 30–32), and the best-layer versus final-layer difference never exceeded $\Delta\rho = 0.009$ (TOST equivalence at margin 0.01 in 7 of 19 tests; the remainder inconclusive, none beyond the margin). Real Foldseek 3Di structure tokens improved SaProt only under leakage-free evaluation ($+0.06$ to $+0.12$, all $p \leq 10^{-4}$), with much smaller gains under the random split ($\leq +0.03$), so both the apparent uselessness of structure tokens and the apparent layer invariance are artifacts of position leakage. A corrected logit lens implementation (fp32, validated against the official logits pathway) showed monotonically improving zero-shot scoring with depth for both models, final layer best (ESM-2 $\rho = 0.443$; SaProt $\rho = 0.437$); the previously reported non-monotonic ESM-2 curve is withdrawn. Multi-scale pooling under the leakage-free protocol occupied a narrow band across four fusion strategies and four readouts (20 datasets per model), with per-fold-trained attention pooling never outperforming the last layer. **Availability and Implementation:** All code and per-dataset results are available at <https://github.com/schwarzeteeguo-creator/esm2-layer-invariance>. An archived snapshot of the code and per-dataset results for this revision is deposited at Zenodo (DOI: 10.5281/zenodo.22844290). The benchmarking pipeline, including masked-position feature extraction, multi-scale pooling, and logit lens analysis, is implemented in Python using PyTorch and scikit-learn.

Keywords protein language models, layer selection, mutation effect prediction, ProteinGym, position leakage, logit lens, CKA

1. Introduction

Protein language models (PLMs) pre-trained via masked language modeling (MLM) on millions of protein sequences have become indispensable for mutation effect prediction, . . . Among these, ESM-2 is one of the most widely adopted, with a 650M-parameter architecture comprising 33 transformer layers that produce 1280-dimensional per-residue hidden states. These

embeddings have been successfully applied to diverse tasks including variant effect prediction, , protein design, and functional annotation, , establishing ESM-2 as a foundational tool in computational biology.

The internal organization of information across transformer layers has been extensively studied in natural language processing, where the “BERTology” literature, has revealed that different layers specialize in distinct linguistic phenomena: lower

layers encode surface-level syntax, middle layers capture semantic content, and final layers refine task-specific predictions. This hierarchical view has motivated analogous investigations in protein language models. A growing body of work posits that different transformer layers in PLMs similarly encode complementary biochemical information, and that combining representations from multiple layers should therefore improve downstream tasks. Kumar and Jha demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, while early layers (6–14) have been shown to encode local physicochemical properties such as hydrophobicity and secondary structure propensity. Alsamkary, Elshaffei et al. proposed hierarchical pooling and pooled attention architectures for fusing representations across multiple PLMs and reported up to a 12% Spearman improvement for protein–protein interaction prediction. More broadly, the strategy of extracting and fusing embeddings from multiple transformer layers has become common practice in PLM-based pipelines, often under the implicit assumption that each layer contributes non-redundant information.

These claims, however, rest on an assumption that has not been systematically tested: that different ESM-2 layers genuinely encode *complementary* information for mutation effect prediction, rather than merely offering different views of the same underlying signal. If layers are functionally similar under a given readout and evaluation protocol, then multi-scale fusion may add cost without improving prediction. If layer complementarity appears only under specific readouts or split protocols, then layer selection should be treated as a conditional design choice rather than a universal rule. Resolving this question carries practical consequences: the computational cost of extracting and storing embeddings from multiple layers is substantial for large-scale screening efforts, and the proliferation of ad-hoc layer-selection heuristics complicates reproducibility.

To resolve this question, we conducted a large-scale benchmark on ProteinGym, a comprehensive collection of deep mutational scanning (DMS) experiments spanning diverse protein families, assay types, and organismal contexts. We employed three independent experimental paradigms (Figure 1): (1) per-layer supervised probing via Ridge regression, (2) multi-scale embedding pooling with controlled feature sets, and (3) logit lens zero-shot scoring. In parallel, we quantified layer-wise representational similarity through cosine similarity analysis. Together, these approaches test when PLM layers behave as practically interchangeable representations for mutation effect prediction and when readout choice or evaluation protocol exposes layer-specific differences. **We report both the artifacts and the leakage-free re-analysis: the random-split layer profile is largely position memorization, while leakage-free evaluation reveals small, consistent late-layer advantages and a measurable benefit of structure tokens that the standard protocol cannot see.**

2. Methods

2.1. Models and Data

We used SaProt.650M.AF2, a structure-aware variant of ESM-2 650M that extends the vocabulary to 446 structure-informed tokens (21 amino acids $\times$ 21 Foldseek 3Di states), as our primary model. For validation, we also tested standard ESM-2

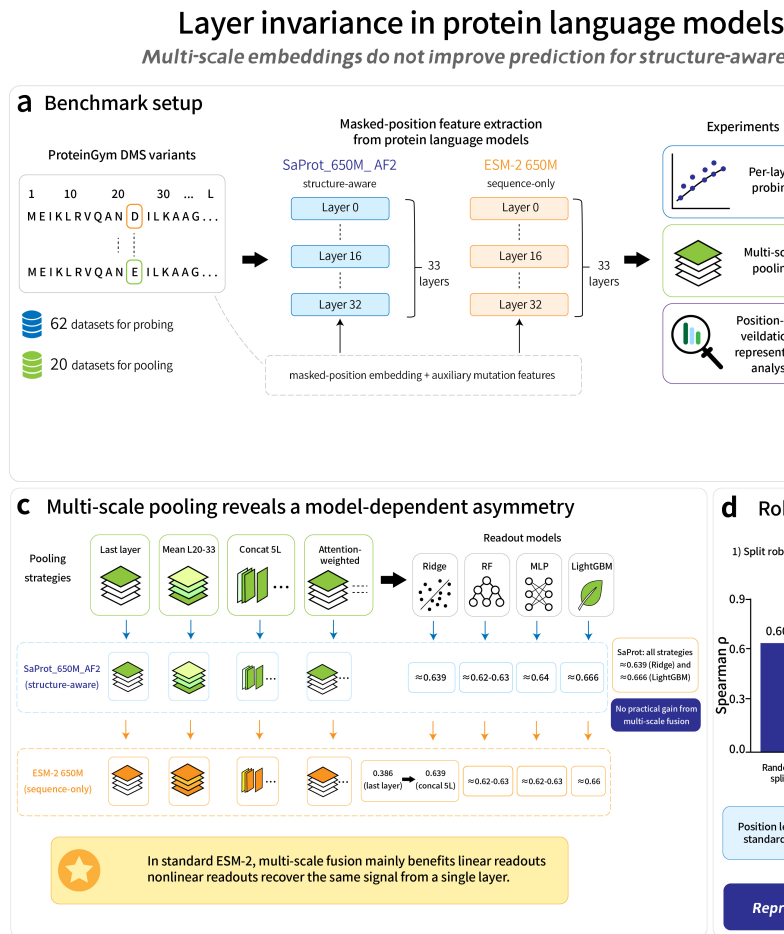


Figure 1 Experimental design overview. The three complementary experimental paradigms employed in this study: (a) Per-layer probing: Ridge regression on hidden states extracted from each of the 33 transformer layers under three feature arms (masked wild-type, unmasked wild-type, and mutant) and four train–test splits (random, modulo, contiguous, and position-grouped), quantifying the predictive information content per layer together with a random-position lookup control. (b) Multi-scale pooling: four embedding fusion strategies (Last Layer, Mean L20–33, Concat 5L, Attention-weighted, with attention weights trained within each cross-validation fold) tested across four readout models (Ridge, Random Forest, MLP, LightGBM) to assess whether combining layers improves prediction. (c) Logit lens: applying the final-layer LM head to intermediate hidden states for zero-shot mutation scoring, directly measuring how token predictions are refined across depth. (d) Layer-wise cosine similarity analysis: quantifying representational divergence independent of predictive performance. Together, these paradigms provide converging evidence on whether different transformer layers encode complementary or redundant information for mutation effect prediction.

650M in both the probing and logit lens experiments, enabling direct comparison between structure-aware and sequence-only representations. All experiments used ProteinGym’s substitution DMS benchmark, comprising 63 datasets with experimentally measured fitness effects for single and multiple amino acid substitutions. All 63 datasets were used for the probing and logit lens experiments ($n = 63$ for the random, modulo, and position-grouped splits; six datasets whose mutated positions form a

single contiguous block have undefined Spearman correlations under the contiguous split, which is therefore reported on the remaining 57). The pooling benchmark was run on a computationally tractable subset of 20 datasets because it crossed four pooling strategies with four readout models and multiple cross-validation folds. We therefore treat the pooling analysis as a controlled validation of the probing result rather than as a full ProteinGym-wide estimate. The subset comprises the first 20 datasets in alphabetical order, fixed identically for both models. Per-dataset mutation counts range from under 100 to over 200,000. For SaProt, we additionally prepared real Foldseek 3Di structure tokens by assigning each wild-type residue the 3Di state of its aligned residue in the AlphaFold Protein Structure Database entry for that sequence (structures available for 62 of 63 datasets; two aligned structures were truncated at the C terminus and the missing residues were padded with the Foldseek gap state; the influenza nucleoprotein dataset has no AFDB entry and was evaluated with structure-masked tokens). SaProt results are reported for both structure-masked and real-3Di inputs.

2.2. Experiment 1: Per-Layer Probing

For each DMS dataset we extracted per-layer hidden states under three feature arms. The **masked_wt** arm replaces each mutated position in the wild-type sequence with the mask token ("M#" for SaProt, <mask> for ESM-2) and reads the hidden state at that position, exactly as in the submitted version. The **unmasked_wt** arm reads the wild-type token at the mutated position without masking. The **mutant** arm feeds the full mutant sequence and reads the hidden state at the mutated position; it is the only arm whose representation differs between two mutations at the same position, and therefore the only one that can carry mutation-specific information beyond position identity. All mutations of a variant were masked jointly in the masked_wt arm. Both SaProt_650M_AF2 (structure-masked and real-3Di inputs) and standard ESM-2 650M were evaluated under identical conditions.

Each arm was evaluated under four train-test splits, all with five cross-validation folds (four for the single four-position dataset under the position-grouped split) and with all rows of a multi-mutant variant assigned to the same fold: the standard **random** split over mutations; the **modulo** split, which assigns each residue position to fold $(\text{position} - 1) \bmod 5$ following the official ProteinGym protocol; the **contiguous** split, which divides the sequence into five contiguous segments and holds out one segment per fold; and a **position-grouped** split (GroupKFold over residue positions, so that all mutations at a given position share a fold). The latter three are leakage-free with respect to positions. For each layer $l \in \{0, \dots, 32\}$ and feature configuration (combined: the hidden state concatenated with 42 auxiliary features, namely one-hot encodings of wild-type and mutant amino acids, the BLOSUM62 substitution score, and a normalized Grantham distance proxy; embedding_only; and auxiliary_only), we trained Ridge regression with the regularization strength selected per dataset by generalized cross-validation (grid 10^{-2} to 10^3), standardization fit inside each training fold, and reported Spearman rank correlation ρ . Datasets were subsampled to at most 10,000 rows before feature construction

and 5,000 rows inside the pipeline (11 of 63 datasets were capped), with a fixed seed shared across arms, splits, and layers. The single dataset with fewer than five mutated positions (four positions) used four position-grouped folds.

To quantify how much probing performance is attributable to position identity alone, we added a **random-position control**: each mutation is represented by a fixed random vector indexed by its residue position (33 independent draws), probed with the identical Ridge protocol with and without auxiliary features. This control contains no sequence information; any performance it recovers measures position memorization.

2.3. Experiment 2: Multi-Scale Pooling

We compared four embedding strategies using fixed auxiliary features, evaluated on the **mutant** arm under the modulo split as the primary leakage-free protocol: (1) **Last Layer**: the standard approach using only the final transformer layer; (2) **Mean L20–33**: mean pooling across layers 20–33, motivated by Kumar and Jha ; (3) **Concat 5L**: concatenation of layers 6, 14, 20, 26, and 32 (0-indexed, including the final layer), spanning the full representational depth; and (4) **Attention-Weighted**: softmax-weighted pooling across all 33 layers, with the weights trained within each cross-validation fold.

For each strategy, we evaluated Ridge regression (regularized by generalized cross-validation as above), Random Forest (50 estimators, max depth 10), MLP (two hidden layers of 256 units, ReLU), and LightGBM (100 estimators, max depth 7) with 3-fold cross-validation for datasets exceeding 3,000 mutations (5-fold otherwise). Standardization was fit inside training folds, and the auxiliary features were identical across strategies, isolating the effect of embedding choice.

2.4. Experiment 3: Logit Lens

We applied the model's final-layer LM head to the hidden states of each intermediate layer, producing pseudo-logits for zero-shot mutation scoring. All computations ran in fp32, and for every dataset the final-layer pseudo-logits were validated against the model's official logits pathway (maximum $|\Delta \log P| < 10^{-3}$; all 63 datasets passed for both models). For standard ESM-2, we used the conventional log-ratio, $\log P(\text{mutant}) - \log P(\text{wild-type})$. For SaProt with structure-masked input tokens (AA#, where # denotes masked structure), we marginalized over all 21 structure-token variants per amino acid; with real 3Di inputs, the wild-type structure token at each position was retained. We reported per-layer Spearman ρ between these zero-shot scores and experimental DMS fitness values across all 63 datasets. The lens curves published in the submitted version were produced by an fp16 pipeline whose numerical error invalidated the ESM-2 curve; they are withdrawn and replaced by these validated results (cf. the tuned lens literature).

2.5. Layer Similarity Analysis

We computed the 33×33 cosine similarity matrix of per-layer hidden states, averaged over 50 randomly selected residue positions across 10 datasets. Because raw cosine is not invariant

to rotations of the coordinate system, we additionally computed linear CKA on the same hidden-state matrices. The unit of analysis for similarity-versus-distance statistics is the set of unique layer pairs (i, j) with $i < j$, diagonal excluded. A high CKA combined with a near-zero cosine indicates a change of basis that preserves linearly decodable information.

2.6. Statistical Evaluation

We used three complementary metrics: (1) **invariance score**, defined as the standard deviation of per-layer mean Spearman ρ (smaller indicates greater invariance); (2) **equivalence testing**: for each model, arm, and leakage-free split, we tested whether the mean per-dataset difference between the best layer and the final layer falls within a margin of $\Delta\rho = 0.01$ using two one-sided tests (TOST) on paired differences, with the best layer fixed as the argmax of the layer-mean curve (19 tests in total), each reported with its 90% bootstrap confidence interval and Cohen's d_z ; and (3) **paired Wilcoxon signed-rank tests** for split contrasts (random versus leakage-free) and for real-3Di versus structure-masked SaProt inputs. We use the term *practical layer invariance* operationally rather than as a formal proof of mathematical equivalence. A layer set was considered practically invariant when per-layer variation was small relative to between-dataset variation, when the best-layer versus final-layer difference had negligible effect size, and when no layer comparison remained significant after false-discovery-rate correction. This framing is intentionally conservative: non-significant hypothesis tests alone are not interpreted as proof that two layers are identical.

3. Results

3.1. Representations Change Dramatically While Predictive Performance Is Preserved

We first asked whether different transformer layers encode representations that are merely different views of the same underlying information, or whether they capture genuinely complementary features. To answer this, we computed the 33×33 layer-wise cosine similarity matrix of per-layer hidden states, averaged over 50 randomly selected residue positions across 10 datasets (Figure 2).

The matrix revealed a smooth representational trajectory across layers. Adjacent layers were highly similar ($\cos = 0.979 \pm 0.011$), and similarity decayed approximately linearly with layer distance (Pearson $r = -0.50$). The decay was smooth and monotonic, with no intermediate trough: for SaProt the mean cosine by distance fell from 0.97 ($d = 2$) through 0.80 ($d = 20$) to 0.50 ($d = 30$), collapsing to -0.009 between the most distant layers (Layer 0 vs. Layer 32). ESM-2 decayed more gently and remained positive throughout (0.93 at $d = 2$, 0.56 at $d = 32$). Crucially, near-zero raw cosine does not imply lost information: the linear CKA between SaProt layers 0 and 32 was 0.808 while their cosine was -0.009 (ESM-2: CKA 0.554, cosine 0.558; adjacent-layer CKA 0.97 for both models). The depth-wise transformation is therefore well described as a rotation

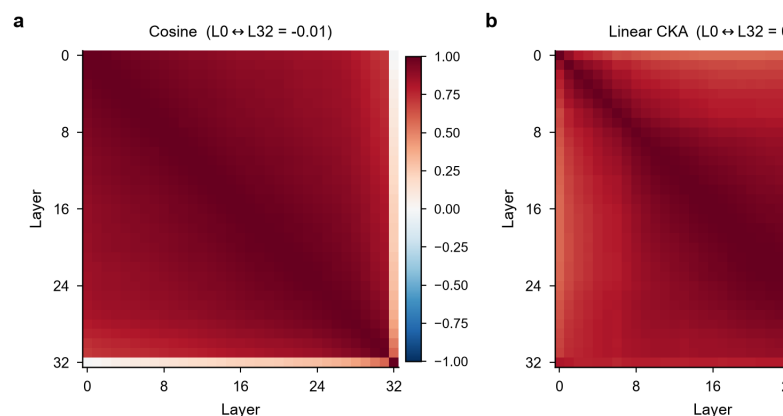


Figure 2 Layer-wise similarity matrices for SaProt (33×33). Averaged over 10 datasets and 50 random residue positions per dataset. (a) Cosine: adjacent layers are highly similar ($\cos = 0.98$); mean similarity decays monotonically with distance and collapses to -0.01 only between the most distant layers (Layer 0 vs. 32). (b) Linear CKA over the same hidden states: the most distant layers retain CKA 0.81, indicating a rotation that preserves linearly decodable information rather than information loss.

of the representation that preserves linearly decodable information, which explains how supervised performance can remain flat across depth while raw cosine collapses (Figure 3c). This large rotation of the representational geometry raises a sharp question: does this representational transformation reflect complementary information encoded at different depths, or is mutation-relevant predictive performance preserved under this transformation?

To answer this, we conducted three converging experimental paradigms, each designed to interrogate the information content of individual layers from a different angle.

3.2. Random-Split Layer Invariance Is Largely Position Memorization

We first asked whether the near-flat random-split layer profile of the submitted version reflects genuine representational redundancy. Under the unified pipeline it replicates exactly: for the masked_wt arm, per-layer mean ρ spans 0.603–0.606 for both models with invariance scores $\sigma \approx 5 \times 10^{-4}$ to 1.2×10^{-3} (Table 2). A Ridge regressor given random vectors indexed only by mutation position plus the same auxiliary features recovered $\rho = 0.600$ under the random split (Table 1), matching the full masked-position PLM probing level. Under leakage-free splits the same control collapsed to $\rho = 0.138$ (modulo), 0.096 (contiguous, $n = 57$), and 0.135 (position-grouped), a paired drop of -0.46 (random versus modulo, Wilcoxon $p = 5.2 \times 10^{-12}$); the random vectors alone carried 0.557 under the random split and approximately zero elsewhere, and the auxiliary features alone fell from 0.339 to 0.19–0.23. Because every mutation at the same position shares one masked-position embedding, the random-split task is solvable by a position lookup table; the flat layer profile is therefore largely an artifact of position memorization rather than evidence that mutation information is distributed uniformly across depth.

Table 1. Random-position lookup control (63 datasets; contiguous $n = 57$). Mean Spearman ρ of Ridge regressors given no sequence information. The control reproduces the random-split PLM probing level and collapses under leakage-free splits.

Features	Random	Modulo	Contiguous
Random position vector + aux	0.600	0.138	0.096
Random position vector only	0.557	−0.008	−0.056
Auxiliary only	0.339	0.230	0.188

Table 2 and Figure 3a report the unified re-probing of all three arms, both models, and all four splits ($n = 63$; contiguous $n = 57$). Two patterns are stable across every leakage-free column. First, the best layer sits in a narrow late band: layers 30–32 are best in 26 of 27 model–arm–split combinations (layer 27 in the remaining one). Second, the best layer is at most marginally better than the final layer: the mean per-dataset difference never exceeded $\Delta\rho = 0.009$, TOST at the 0.01 margin declared equivalence in 7 of 19 leakage-free tests, and the remaining 12 were inconclusive (90% confidence intervals overlapping the margin); no test indicated a difference beyond the margin (Supplementary Table S4). The invariance score rose from $\sigma \approx 10^{-3}$ (masked_wt, random) to $4\text{--}7 \times 10^{-2}$ under leakage-free evaluation, yet remained well below the cross-dataset standard deviation (≈ 0.17): performance varies far more across datasets than across layers, but under leakage-free evaluation the residual layer variation has a consistent direction, favoring the final layers. The mutant arm exceeded the masked arms in every split (e.g., 0.681 versus 0.606 under random, 0.483 versus 0.459 under modulo for ESM-2), confirming that mutation-specific signal exists and is recoverable once the representation actually depends on the mutation.

Table 2. Unified per-layer probing, combined features (mean Spearman ρ ; best layer in parentheses, followed by the final-layer value when the best layer is not layer 32). $n = 63$; contiguous $n = 57$. Full 33-layer curves in Supplementary Table S2; per-dataset results in Supplementary Table S3.

Model	Arm	Random	Modulo	Contiguous	Pos.-grouped
ESM-2	masked_wt	0.606 (32)	0.459 (32)	0.370 (32)	0.450 (32)
ESM-2	unmasked_wt	0.606 (32)	0.470 (32)	0.388 (32)	0.461 (32)
ESM-2	mutant	0.681 (32)	0.483 (31)	0.479 (31)	0.475 (31)
SaProt (mask)	masked_wt	0.603 (31) → 0.603	0.391 (30)	0.386 (31) → 0.274	0.385 (31) → 0.382
SaProt (mask)	unmasked_wt	0.604 (32)	0.412 (30)	0.406 (31) → 0.324 (31)	0.320 (27) → 0.400
SaProt (mask)	mutant	0.652 (23) → 0.651	0.465 (30)	0.460 (32) → 0.323 (32)	0.461 (30) → 0.458
SaProt (3Di)	masked_wt	0.606 (32)	0.461 (31)	0.460 (31) → 0.394 (31)	0.454 (31) → 0.453
SaProt (3Di)	unmasked_wt	0.606 (27) → 0.605	0.473 (31)	0.466 (31) → 0.415 (31)	0.467 (31) → 0.459
SaProt (3Di)	mutant	0.684 (26) → 0.683	0.531 (31)	0.525 (31) → 0.451 (31)	0.445 (32) → 0.524

3.3. Multi-Scale Pooling Shows No Practical Gain Under a Leakage-Free Protocol

We re-ran the pooling benchmark with the unified pipeline (mutant arm, modulo split, GCV-regularized Ridge, fold-internal

standardization, attention weights trained per fold) on the 20-dataset design subset for each model (Table 3; Figure 3b). For SaProt (structure-masked inputs), all four strategies occupied a narrow band for every readout (Ridge spread across strategies 0.005–LightGBM 0.014). For ESM-2, spreads were likewise small (Ridge 0.041; LightGBM 0.050), and the best single configuration was the last layer with LightGBM ($\rho = 0.508$), exceeding every multi-scale strategy under every readout. Per-fold-trained attention pooling was never the best strategy for either model.

Table 3. Multi-scale pooling under the unified leakage-free pipeline (mutant arm, modulo split, combined features; 20 datasets per model; mean $\rho \pm$ sd). Spreads across strategies: ESM-2 Ridge 0.041, RF 0.057, MLP 0.045, LightGBM 0.050; SaProt Ridge 0.005, RF 0.010, MLP 0.039, LightGBM 0.014.

ESM-2 650M	Ridge	RF	MLP
Last Layer (baseline)	0.463 $\pm$ 0.174	0.476 $\pm$ 0.173	0.480 $\pm$ 0.198
Mean L20–33	0.473 $\pm$ 0.180	0.420 $\pm$ 0.165	0.436 $\pm$ 0.179
Concat 5L	0.481 $\pm$ 0.183	0.453 $\pm$ 0.181	0.435 $\pm$ 0.170
Attention-weighted	0.439 $\pm$ 0.201	0.419 $\pm$ 0.209	0.440 $\pm$ 0.209
SaProt 650M_AF2	Ridge	RF	MLP
Last Layer (baseline)	0.432 $\pm$ 0.163	0.415 $\pm$ 0.174	0.445 $\pm$ 0.168
Mean L20–33	0.433 $\pm$ 0.166	0.415 $\pm$ 0.166	0.429 $\pm$ 0.166
Concat 5L	0.434 $\pm$ 0.185	0.407 $\pm$ 0.173	0.406 $\pm$ 0.176
Attention-weighted	0.429 $\pm$ 0.164	0.405 $\pm$ 0.166	0.426 $\pm$ 0.164

3.3.1. The Previously Reported ESM-2 Asymmetry Was a Cross-Protocol Artifact

The large ESM-2 asymmetry in the submitted version (last-layer Ridge $\rho = 0.386$ versus Concat 5L $\rho = 0.639$) does not survive protocol unification. It arose from comparing feature constructions that differed in masking state, regularization, and split protocol. Under the unified leakage-free pipeline the corresponding contrast is 0.463 versus 0.481 (+0.018), inside the narrow band described above. The MLP readout, previously reported at $\rho \approx 0.33$ across strategies, reflected an under-trained surrogate configuration (the submitted pipeline had one hidden layer of 28 units, 50.3% dropout) rather than the described architecture, with the described two-layer architecture performing on par with Ridge here. We therefore withdraw the claim that standard ESM-2 exhibits a large readout-specific layer dependence; residual readout interactions are an order of magnitude smaller under the unified protocol, and no strategy consistently beats the last layer.

3.4. Logit Lens Confirms Late-Layer Information Accumulation

The re-implemented lens (fp32, per-dataset validation against the official logits pathway, $n = 63$) produced monotonically improving zero-shot scores with depth for both models (Figure 4). ESM-2 rose from $\rho = 0.031$ (layer 0) to $\rho = 0.443$ (layer 32), with only shallow fluctuations in early layers; SaProt with real 3Di tokens rose from -0.077 (layer 0) through approximately zero around layer 20 to $\rho = 0.437$ (layer 32). Both curves

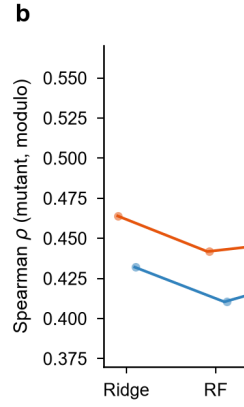
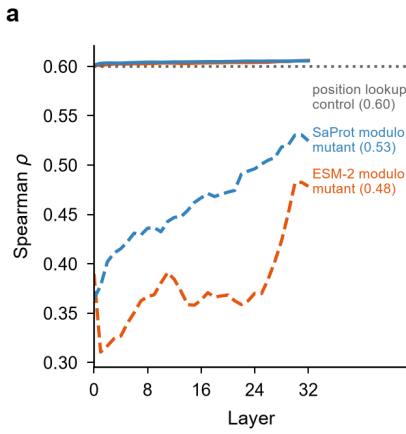


Figure 3 Per-layer probing, multi-scale pooling, and layer distance under the unified protocol. (a) 33-layer probing (combined features): under the random split the masked-arm profile is flat for both models ($\sigma \approx 10^{-3}$) and is reproduced by the random-position lookup control (dotted, 0.600); under leakage-free splits (modulo, mutant arm) performance is lower and the best layers are consistently late (layers 30–32, $\Delta\rho \leq 0.009$ versus the final layer). (b) Pooling (mutant arm, modulo split, 20 datasets per model; four strategies as dots, mean connected): ESM-2 spread across strategies 0.041 (Ridge) to 0.057 (RF); SaProt spread 0.005 (Ridge) to 0.039 (MLP); last-layer LightGBM is the best single configuration; per-fold-trained attention pooling is never best. (c) Cosine similarity versus layer distance (unique layer pairs): SaProt decays monotonically and reaches -0.01 only at the largest distances, while ESM-2 remains positive; linear CKA between the most distant SaProt layers remains 0.81 (Figure 2), indicating rotation with preserved linear information rather than loss.

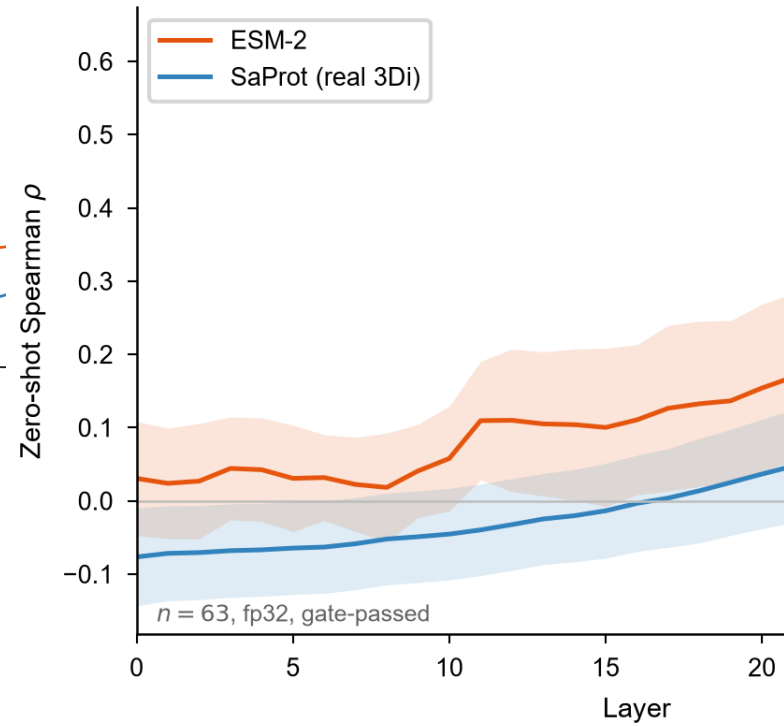


Figure 4 Logit lens: zero-shot mutation scoring by layer ($n = 63$). Both models improve with depth and attain their maximum at the final layer: ESM-2 (orange) rises from $\rho = 0.03$ (layer 0) to $\rho = 0.443$ (layer 32); SaProt with real 3Di tokens (blue) rises from negative early-layer values to $\rho = 0.437$ (layer 32). Shaded bands indicate ± 1 standard deviation across datasets. All computations in fp32 and validated against the official logits pathway; the previously reported non-monotonic ESM-2 curve is withdrawn.

attain their maximum at the final layer, at magnitudes consistent with published zero-shot PLM scoring. The submitted version instead reported a non-monotonic ESM-2 curve peaking at layer 1 ($\rho = 0.056$) and ending at -0.021 ; that curve was produced by an fp16 pipeline and is withdrawn. The corrected result removes the apparent model contrast and agrees with the leakage-free probing: mutation-relevant information accumulates toward the final layers. Zero-shot absolute levels remain below supervised probing (0.44 versus up to 0.53), so the lens is best read as a consistency check on layer-wise trends rather than a standalone benchmark.

3.5. Boundary Conditions

The results above provide evidence for practical layer invariance under common random-split supervised benchmarking conditions. We now examine the two factors that moderate this conclusion: evaluation protocol and input representation.

3.5.1. Leakage-Free Splits: What Is Lost and What Remains

The control and main-table results above already localize the leakage: the random-position lookup table reproduces the random-split level (0.600 versus 0.603–0.606 for full probing) and loses it entirely under leakage-free splits (0.10–0.14), while the random vectors alone carry 0.557 under the random split and approximately zero elsewhere (Tables 1 and 2; Figure 5). Among

the leakage-free splits, modulo and position-grouped give similar levels (e.g., SaProt real-3Di mutant arm: 0.531 and 0.532) while contiguous is uniformly hardest (0.451), consistent with sequence-contiguous segments correlating with local structural context and mutational scope. Random mutation splitting, the de facto standard in DMS benchmarking, therefore inflates absolute performance estimates by roughly 0.15–0.20 ρ in these arms and masks both layer differences and structure-token contributions. We recommend that future DMS benchmarks report at least one leakage-free split alongside the standard random split, analogous to the time-series community's practice of reporting both random and temporal cross-validation.

Together, these boundary conditions refine the central finding: apparent layer invariance under random splitting is dominated by position memorization, leakage-free evaluation reveals small but consistent late-layer advantages, and the benefit of structure tokens is itself conditional on the evaluation protocol.

3.6. Structure Tokens Help Only Under Leakage-Free Evaluation

Because SaProt can be run with real 3Di tokens or with structure-masked tokens, the unified protocol allows a paired test, dataset by dataset and layer by layer, of what structure tokens actually contribute (Table 4). Under the random split the

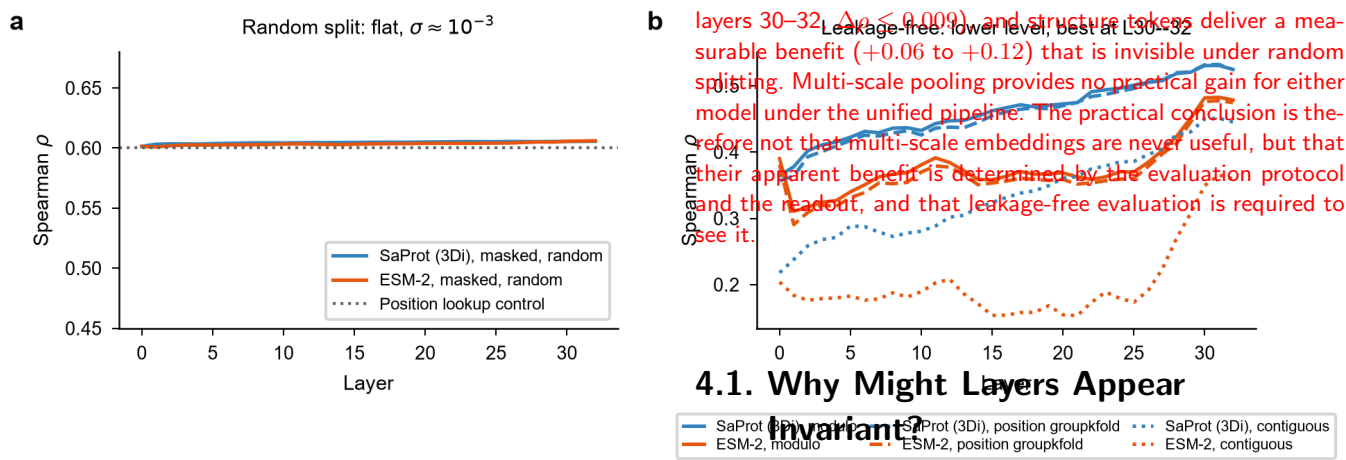


Figure 5 Random versus leakage-free split probing (SaProt, 63 datasets). (a) Random mutation split: flat per-layer profile ($\sigma \approx 10^{-3}$), reproduced by the random-position lookup control ($\rho = 0.600$). (b) Leakage-free splits (modulo, position-grouped, contiguous): performance drops to 0.27–0.53 depending on arm and model, the invariance score rises to $4-7 \times 10^{-2}$, and the best layers shift consistently to layers 30–32.

two input modes were nearly identical (paired differences at layer 32: +0.003 masked_wt, +0.002 unmasked_wt, +0.032 mutant). Under every leakage-free split the real-3Di inputs were consistently superior, with mean paired advantages of +0.059 to +0.122 across arms and splits, all with Wilcoxon $p \leq 10^{-4}$. Structure tokens therefore carry genuine mutation-relevant information, but that contribution becomes visible only when position memorization is removed from the evaluation. This conditional benefit also retroactively explains why structure-aware inputs appeared useless in random-split comparisons: the position channel, available to both input modes, saturates the task.

Table 4. Real-3Di versus structure-masked SaProt inputs (paired per-dataset differences at layer 32, combined features; Wilcoxon signed-rank p). The benefit of structure tokens is small and mostly non-significant under the random split and large and highly significant under all leakage-free splits.

Arm	Random		Modulo	Contiguous	Position groupfold
	$\Delta\rho$	p			
masked_wt	+0.003	2.2×10^{-2}	+0.073 (3.5×10^{-7})	+0.126 (8.7×10^{-6})	+0.011 (6.1×10^{-2})
unmasked_wt	+0.002	9.1×10^{-1}	+0.060 (4.6×10^{-6})	+0.080 (2.1×10^{-5})	+0.055 (1.2×10^{-3})
mutant	+0.032	2.5×10^{-6}	+0.064 (6.9×10^{-5})	+0.122 (0.3×10^{-5})	+0.067 (6.0×10^{-5})

4. Discussion

The central result of this study is conditional rather than absolute. Under the standard random-split protocol, the 33 transformer layers of SaProt and ESM-2 show near-flat mutation effect prediction performance, but the position-lookup control shows that this flatness is largely position memorization rather than distributed mutation information. Under leakage-free evaluation, genuine mutation-specific performance is lower (ρ up to 0.53), small but consistent late-layer advantages emerge (best

layers 30–32, $\Delta\rho \leq 0.009$), and structure tokens deliver a measurable benefit (+0.06 to +0.12) that is invisible under random splitting. Multi-scale pooling provides no practical gain for either model under the unified pipeline. The practical conclusion is therefore not that multi-scale embeddings are never useful, but that their apparent benefit is determined by the evaluation protocol and the readout, and that leakage-free evaluation is required to see it.

4.1. Why Might Layers Appear Invariant?

We propose that this practical invariance may be a consequence of the MLM pre-training objective. During pre-training, all layers receive gradients through the same cross-entropy loss computed at the final LM head. While this setup allows layers to develop distinct representational formats (as the cosine similarity analysis shows), it may not strongly incentivize each depth to specialize in fundamentally different mutation-relevant factors. Consequently, a supervised readout trained on individual layers can recover very similar performance under random-split evaluation, where position memorization dominates the task.

The cosine similarity between layers decays monotonically with distance and, for SaProt, collapses to approximately zero only between the most distant layers, while the linear CKA between the same layers remains 0.81. This combination, a near-rotation that preserves linearly decodable information, is consistent with the effect of residual connections: because residual streams accumulate representations from all preceding layers, late layers retain a linearly accessible trace of early-layer content even as their raw coordinates rotate. The U-shaped recovery described in the submitted version is not present in the recomputed curves and is withdrawn.

This interpretation is consistent with the logit lens hypothesis, which posits that intermediate layers in transformer language models progressively refine token predictions toward the final output distribution, rather than constructing qualitatively different representations. The corrected lens curves support this directly for both models: zero-shot scoring improves with depth and peaks at the final layer. The near-orthogonality of the most distant SaProt layers (≈ -0.01) alongside their high linear CKA (0.81) combined with near-flat supervised probing performance suggests that mutation-relevant signal may remain accessible while representations move through successive coordinate frames.

Notably, this pattern contrasts with findings from NLP transformers, where probing classifiers consistently recover different linguistic features from different layer depths. In BERT, for example, surface syntactic features peak in early layers while semantic features dominate in middle-to-late layers. The relative flatness of the PLM probing curve may reflect the fundamentally different nature of the learning signal in protein versus natural language: whereas language contains hierarchical structure at multiple timescales (morphology, syntax, semantics, discourse), the MLM task on protein sequences operates on a single level of abstraction (amino acid identity), potentially eliminating the pressure for layer-wise specialization.

4.1.1. What Separates SaProt from Standard ESM-2?

Under the unified protocol the two models differ mainly in two measured ways: how much their final representations retain early-layer geometry (ESM-2 distant-layer cosine 0.56 with CKA 0.55; SaProt cosine -0.01 with CKA 0.81), and the benefit conferred by structure tokens. The submitted version interpreted the pooling contrast between the models as structure-aware training distributing information more uniformly across layers; that contrast dissolved under protocol unification, and we no longer draw that inference. The surviving, directly measured difference is the conditional structure-token benefit: real 3Di inputs improve leakage-free probing by $+0.06$ to $+0.12$ while leaving random-split performance essentially unchanged. Structure information is genuinely present and genuinely useful, but only leakage-free evaluation can see it.

This framing carries a testable prediction: fine-tuning standard ESM-2 with structural objectives (e.g., contact prediction or fold classification) should reproduce a similar leakage-free advantage over the sequence-only baseline. This remains a mechanistic hypothesis rather than a direct result of the present experiments.

4.2. Reconciling with Prior Layer-Selection Studies

Our findings do not contradict the observation that different layers may be optimal for different task types. Kumar and Jha demonstrated that mid-to-late ESM-2 layers (20–33) outperform the final layer for kinase function prediction, achieving 32% higher clustering quality and superior classification accuracy. This is compatible with our narrower conclusion: different downstream tasks may induce different optimal readout directions, while mutation effect prediction under random-split probing shows little layer dependence. The key distinction is between task-level specialization and within-task layer complementarity under a specified readout and evaluation protocol.

Similarly, the 12% Spearman improvement reported by Alsamkary, Elshaffei et al. for attention-based fusion architectures was observed in the context of multi-chain protein–protein interaction prediction, a task that differs substantially from per-residue mutation effect scoring. The benefit of sophisticated pooling may therefore be task-dependent rather than inherent to multi-scale representation fusion. For mutation effect prediction, our results suggest that such improvements should be interpreted in light of model type, readout choice, and split protocol.

4.3. Practical Implications

For researchers using protein language model embeddings for mutation effect prediction, our results support two practical recommendations that depend on the model variant, readout model, and evaluation protocol:

- 1. **Structure-aware models (SaProt, ESM-2 fine-tuned with structural objectives):** The final transformer layer is a strong default. In our benchmarks, multi-scale fusion provided no practical benefit over single-layer embeddings for either model under leakage-free evaluation, and 1280

dimensions sufficed for both linear and nonlinear readouts. This reduces storage and computational costs by a factor of 5–10 compared to the common practice of concatenating 6–8 layers (up to 10,240 dimensions).

- 2. **Standard PLMs (ESM-2, ProtBERT, etc.):** Under leakage-free evaluation, concatenation of 3–5 strategically spaced layers offers at most a marginal linear-readout gain ($+0.018$ in our experiments, from $\rho = 0.463$ to 0.481). Switching the readout matters more than fusing layers: last-layer LightGBM reached $\rho = 0.508$, the best pooling configuration overall, achieving the benefits claimed for multi-scale fusion without the dimensionality cost.

Table 5 quantifies the practical trade-offs. Concat 5L uses $4.9\times$ more features than single-layer approaches and incurs roughly $25\times$ the computational cost for Ridge regression (the covariance matrix grows from $1,322^2$ to $6,442^2$ entries), yet offered no benefit on either model in our benchmark and was matched or exceeded on ESM-2 by switching from Ridge to LightGBM on a single layer. Storing Concat 5L embeddings for a typical DMS benchmark ($\sim 10,000$ variants) requires 258 MB versus 53 MB for a single layer, a gap that widens linearly with dataset size and becomes substantial for genome-scale studies.

Table 5. Computational cost comparison across pooling strategies. “Per-mutation” storage assumes float32 precision. Time estimates are per-dataset averages on an NVIDIA RTX 5090, including embedding extraction and readout training across 20 ProteinGym DMS subsets. Best achievable ρ from Table 3 (mutant arm, modulo split).

	Last Layer	Mean L20–33	Att
Feature dimension	1,322	1,322	1,
Storage per mutation (KB)	5.2	5.2	5
Ridge design matrix (5,000 vars)	6.6×10^6	same	sa
Ridge covariance matrix entries	1.7×10^6	same	sa
Avg. time per dataset (min)	~ 3	~ 3	$\sim$
Best achievable ρ (ESM-2 / SaProt):			
With Ridge	0.463 / 0.432	0.473 / 0.433	0.439
With LightGBM	0.508 / 0.448	0.457 / 0.439	0.463

In both cases, computational resources may be better directed toward feature types that are plausibly complementary to PLM embeddings:

- 1. Structural features (e.g., AlphaFold-predicted pLDDT, PAE , or Foldseek 3Di states), which complement sequence-based embeddings;
- 2. Sequence-level representations that capture global protein properties beyond per-residue features;
- 3. Multiple sequence alignment (MSA)-based approaches , that leverage evolutionary information beyond what PLMs encode.

As discussed under Boundary Conditions (Section 3.5), the position-lookup control shows that standard evaluation protocols can conflate mutation-effect learning with position-level memorization. We recommend that future DMS benchmarks adopt position-aware evaluation as a standard complement to random splitting.

4.4. Limitations

Several limitations should be noted. First, we tested the 650M-parameter ESM-2 variant and its structure-aware derivative SaProt. Smaller (35M, 150M) and larger (3B, 15B) variants may show different degrees of layer specialization, and this remains to be tested. Second, our experiments are restricted to per-residue mutation effect prediction; for other tasks such as kinase function prediction, subcellular localization, or protein-protein interaction scoring, layer-wise differences and the benefits of multi-scale fusion may be more pronounced. Third, our pooling benchmark used a 20-dataset validation subset, which is smaller than the 63-dataset probing analysis. Fourth, the SaProt pooling runs use structure-masked inputs, whereas the probing and logit lens runs include real 3Di inputs, and one dataset (influenza nucleoprotein) lacks an AFDB structure. Fifth, our feature ablation (embedding-only vs. auxiliary-only) indicates that learned embeddings dominate predictive performance in this setting, but it does not rule out the possibility that nonlinear combinations of auxiliary features could provide stronger baselines. Sixth, the maximum sample size of 10,000 mutations per dataset (subsampled to 5,000 for regression; 11 of 63 datasets were capped) may limit sensitivity to subtle layer-wise effects on the largest DMS screens (>200,000 mutations), though the consistency of invariance scores across dataset-size categories argues against this concern. Seventh, our linear probing approach can only detect information that is linearly separable; nonlinear interactions between layers could theoretically exist but would require a different experimental design to identify. Eighth, position-level data splitting (Section 3.5) shows that random mutation splits permit position-level leakage, which the lookup control quantifies and which now motivates our reporting of leakage-free splits throughout; future DMS benchmarks should adopt position-aware evaluation as a standard protocol alongside random splitting. Finally, our equivalence margin ($\Delta\rho = 0.01$) is a reporting convention rather than a task-derived requirement, and our statistical analysis supports practical invariance by small effect sizes and relative variation, but it does not constitute a pre-specified formal equivalence test.

4.5. Conclusion

Under leakage-free evaluation, protein language model layers are close to interchangeable for mutation effect prediction, with a small consistent final-layer advantage; under the standard random split, apparent interchangeability is dominated by position memorization. Under standard random-split probing, SaProt and ESM-2 showed near-flat per-layer performance despite large changes in representational geometry. Multi-scale pooling added no practical benefit over the final layer for either model under the unified leakage-free pipeline. Structure tokens helped SaProt, but only under leakage-free evaluation. Position-level evaluation showed that random splits inflate both absolute performance and apparent invariance. These findings argue for a more empirical use of PLM layers: choose the layer strategy jointly with the readout model and the evaluation protocol, and treat claims of layer-specific representations in PLMs with the same scrutiny that has become standard in the NLP community.

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Competing Interests

The authors declare no competing interests.

Data Availability

Code and results are available at <https://github.com/schwarzeteeguo-creator/esm2-layer-invariance>. An archived snapshot of the code, per-dataset results, and extracted structure tokens for this revision is deposited at Zenodo (concept DOI 10.5281/zenodo.22844290; archived version v2.0.1, DOI 10.5281/zenodo.22844315). Raw ProteinGym data: <https://github.com/OATML-Markslab/ProteinGym>.

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